Supplemental Material for “Winding-Free Exact Diagonalization of Quantum Spin Ice”

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This Supplemental Material provides the FCC-32 geometry, perturbative row construction, transported-dipole projection, exact mask benchmark, thermodynamic and spectral definitions, and the $Ce_2Hf_2O_7$ parameter constraint used in the Letter. Every many-body result reported here and in the Letter is obtained on the same 32-site, 2,970-state ice manifold. No smaller-cluster spectrum or interpolation between finite-cluster momentum sectors enters the reported results.

FCC-32 CLUSTER AND ICE MANIFOLD

We use the pyrochlore basis, in units of the conventional cubic lattice constant,

$$\boldsymbol{r}_0 = (0,0,0), \quad \boldsymbol{r}_1 = (0,\tfrac{1}{4},\tfrac{1}{4}), \quad \boldsymbol{r}_2 = (\tfrac{1}{4},0,\tfrac{1}{4}), \quad \boldsymbol{r}_3 = (\tfrac{1}{4},\tfrac{1}{4},0), \tag{1}$$

and a $2 \times 2 \times 2$ array of primitive FCC cells generated by

$$\boldsymbol{a}_1 = (\tfrac{1}{2},\tfrac{1}{2},0), \quad \boldsymbol{a}_2 = (\tfrac{1}{2},0,\tfrac{1}{2}), \quad \boldsymbol{a}_3 = (0,\tfrac{1}{2},\tfrac{1}{2}). \tag{2}$$

Equivalently, the simulation-cell rows are

$$\boldsymbol{L}_1 = (1,1,0), \qquad \boldsymbol{L}_2 = (1,0,1), \qquad \boldsymbol{L}_3 = (0,1,1). \tag{3}$$

The cluster has 32 spins, 96 nearest-neighbor bonds, and 16 tetrahedra. For each directed bond (i,j), the code stores an image vector $\boldsymbol{n}_{ij} \in \mathbb{Z}^3$ such that

$$\boldsymbol{d}_{ij} = \boldsymbol{r}_i - \boldsymbol{r}_j + \boldsymbol{n}_{ij}\boldsymbol{L} \tag{4}$$

is the directed minimum-image displacement; reversing the bond changes the sign of both $\boldsymbol{d}_{ij}$ and $\boldsymbol{n}_{ij}$.

For a spin configuration s, define the tetrahedron charge $Q_t = \sum_{i\in t} S_i^z$. Up to a constant,

$$H_0 = \frac{J_{zz}}{2}\sum_t Q_t^2. \tag{5}$$

The ice basis $\mathcal{I}$ contains the 2,970 bit configurations satisfying $Q_t = 0$ on every tetrahedron. Independent cycle enumeration on the periodic graph gives the census in Table I. A loop is classified by unwrapping its stored image vectors; no visual or home-cell criterion is used.

EFFECTIVE ROWS THROUGH THIRD ORDER

Write $H = H_0 + V$ with

$$V = -J_\pm \sum_{\langle ij\rangle}(S_i^+ S_j^- + S_i^- S_j^+), \tag{6}$$

TABLE I. FCC-32 cycle census. Each entry is contractible plus wrapping.

sites	bonds	tetrahedra	ice states	four-cycles	hexagons
32	96	16	2,970	$0+24$	$32+96$

and let P project onto $\mathcal{I}$, $Q = 1 - P$. One exchange creates a pair of charges and therefore $PVP = 0$. With $R = -QH_0^{-1}Q$, the off-diagonal operators required here are

$$H_{\text{eff}}^{(2)} = PVRVP, \qquad H_{\text{eff}}^{(3)} = PVRVRVP. \tag{7}$$

The implementation constructs these matrices row by row. From every ice state it applies every allowed directed bond exchange, evaluates the exact intermediate Ising energy from Eq. (5), and repeats until the path returns to ice. Source, target, perturbative coefficient, and transported image data are retained before equivalent rows are combined.

Each first hop costs J_{zz}. A flippable four-loop has four orderings that return to ice after the second hop, giving

$$g_4 = 4J_{\pm}^2/J_{zz}. \tag{8}$$

Because Table I contains no contractible four-cycle, every such FCC-32 matrix element is a torus artifact. A flippable hexagon has two perfect matchings and six orderings for each matching, giving the physical ring amplitude

$$g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2 \tag{9}$$

and $g_4/g_{\text{hex}} = J_{zz}/(3|J_{\pm}|)$. The row construction reproduces these coefficients directly; neither amplitude is inserted by hand.

For a dipolar–octupolar nearest-neighbor model, diagonalizing the x–z exchange block by a rotation about y gives principal axes $\widetilde{\tau}$ and angle

$$\tan(2\theta) = \frac{2J_{xz}}{J_x - J_z}, \qquad \tau^z = \cos\theta\,\widetilde{\tau}^z + \sin\theta\,\widetilde{\tau}^x. \tag{10}$$

For the $\widetilde{J}_x$-dominant branch used in the Letter, choose gauge coordinates

$$S^z = \widetilde{\tau}^x, \qquad S^x = \widetilde{\tau}^z, \qquad S^y = -\widetilde{\tau}^y. \tag{11}$$

The $U(1)$ reduction is then

$$J_{zz} = \widetilde{J}_x, \qquad J_{\pm} = -\frac{\widetilde{J}_y + \widetilde{J}_z}{4}, \tag{12}$$

while the physical local dipole is

$$\tau^z = \cos\theta\,S^x + \sin\theta\,S^z = \frac{\cos\theta}{2}(S^+ + S^-) + \sin\theta S^z. \tag{13}$$

TRANSPORTED-DIPOLE PROJECTION

Consider a complete virtual row $r : s \to t$. Let R_r and L_r be the sites raised and lowered between its endpoint ice states, and define the home-cell endpoint polarization change

$$\boldsymbol{\rho}_{ts} = \sum_{i\in R_r} \boldsymbol{r}_i - \sum_{j\in L_r} \boldsymbol{r}_j. \tag{14}$$

If its directed hops accumulate image vector $\boldsymbol{N}_r = \sum_{\ell\in r} \boldsymbol{n}_\ell$, its transported dipole is

$$\boldsymbol{\delta}_r = \boldsymbol{\rho}_{ts} + \boldsymbol{N}_r \boldsymbol{L}. \tag{15}$$

The sign follows the directed-bond convention above. Changing which image is called the home cell shifts the two terms oppositely, so their sum is invariant.

The same result follows by coupling every exchange to a uniform vector potential. A path acquires

$$\prod_{\ell\in r} V_\ell(\boldsymbol{A}) = \exp[-i\boldsymbol{A}\cdot(\boldsymbol{\rho}_{ts} + \boldsymbol{N}_r\boldsymbol{L})] \prod_{\ell\in r} V_\ell(0) \tag{16}$$

$$= e^{-i\boldsymbol{A}\cdot\boldsymbol{\delta}_r} \prod_{\ell\in r} V_\ell(0). \tag{17}$$

Thus the zero Fourier component is the operator-level projection

$$\Pi_0[H_{\text{eff}}] = \sum_{r:\boldsymbol{\delta}_r=0} c_r |t_r\rangle\langle s_r|. \tag{18}$$

It acts before diagonalization and is therefore distinct from averaging spectra or observables obtained at different boundary conditions.

The connection to loop winding can be made explicit. Write an elementary alternating loop as $C = (i_0, i_1, \ldots, i_{2m-1})$ and unwrap its coordinates so that

$$\boldsymbol{R}_{k+1} - \boldsymbol{R}_k = \boldsymbol{r}_{i_{k+1}} - \boldsymbol{r}_{i_k} + \boldsymbol{n}_k \boldsymbol{L}, \qquad \boldsymbol{R}_{2m} = \boldsymbol{R}_0 + \boldsymbol{w}_C \boldsymbol{L}. \tag{19}$$

For a fixed endpoint spin flip, the two perfect matchings transport the initially occupied alternating sites to their two neighboring empty sites. Their transported dipoles can be written

$$\boldsymbol{\delta}_C^{(0)} = \sum_{p=0}^{m-1} (\boldsymbol{R}_{2p+1} - \boldsymbol{R}_{2p}), \tag{20}$$

$$\boldsymbol{\delta}_C^{(1)} = \sum_{p=0}^{m-1} (\boldsymbol{R}_{2p-1} - \boldsymbol{R}_{2p}) = \boldsymbol{\delta}_C^{(0)} - \boldsymbol{w}_C \boldsymbol{L}, \tag{21}$$

where $\boldsymbol{R}_{-1} = \boldsymbol{R}_{2m-1} - \boldsymbol{w}_C \boldsymbol{L}$. Reversing the endpoint spin orientation reverses both vectors. Substitution of the pyrochlore bond vectors for every elementary four- and six-cycle on FCC-32 gives

$$\boldsymbol{\delta}_C^{(0)} = \frac{1}{2}\boldsymbol{w}_C \boldsymbol{L}, \qquad \boldsymbol{\delta}_C^{(1)} = -\frac{1}{2}\boldsymbol{w}_C \boldsymbol{L}. \tag{22}$$

Thus both matchings have zero transport for a contractible loop and nonzero transport for a wrapping loop. Equation (22) is also checked row by row rather than assumed from the cycle label. The projection deletes the 24 four-loops and 96 wrapping hexagons while retaining all 32 contractible hexagons, every virtual ordering, and therefore the complete coefficient g_{hex}.

Ordinary boundary twists couple only to $\boldsymbol{N}_r$. One matching of a wrapping four-loop can consist entirely of home-cell bonds and have $\boldsymbol{N}_r = 0$, even though the endpoint polarization reveals nonzero transport. An eight-corner boundary-twist average consequently retains one half of the absolute matrix element in each wrapping channel. It is not Eq. (18).

POLARIZATION-PARITY MASK AND EXACT FCC-32 TEST

Define the diagonal polarization operator

$$\hat{\boldsymbol{X}} = \sum_i \boldsymbol{r}_i S_i^z, \qquad \boldsymbol{x}_\alpha = \langle \alpha | \hat{\boldsymbol{X}} | \alpha \rangle. \tag{23}$$

Conjugating an exchange by $V(\boldsymbol{\theta}) = e^{2i\boldsymbol{\theta}\cdot\hat{\boldsymbol{X}}}$ supplies its home-cell displacement character. Combining that character with the corresponding boundary image character yields the complete transported-dipole phase. On the minimal $\mathbb{Z}_2^3$ grid, character averaging gives

$$\overline{H}_{\alpha\beta} = H_{\alpha\beta} \prod_{a=1}^{3} \frac{1 + e^{2i\pi(x_{\alpha a} - x_{\beta a})}}{2} \tag{24}$$

$$= H_{\alpha\beta}\big[2(\boldsymbol{x}_\alpha - \boldsymbol{x}_\beta) \text{ is componentwise even}\big]. \tag{25}$$

This is the polarization-parity mask used in the Letter.

At $J_\pm/J_{zz} = -0.05$, the FCC-32 periodic matrix has 91,482 entries larger than $10^{-14} J_{zz}$; both explicit row filtering by $\boldsymbol{\delta} = 0$ and Eq. (25) retain 18,330. Their maximum entrywise residual is exactly zero in double precision. The equality is also channel-resolved: the retained fractions of the wrapping four-loop, wrapping hexagon, and contractible hexagon amplitudes are respectively 0, 0, and 1.

The $\mathbb{Z}_2^3$ grid resolves transport parity. On FCC-32 every wrapping process through third order is odd, so Eq. (25) is identical to the exact zero-transport row projector at the order used in this work. At higher order, an even nonzero winding can occur and a finer character grid is required. We make no all-order claim for the minimal mask.

EXECUTABLE PROTOCOL AND SCOPE

The calculation used for every Letter result follows the same operator-level sequence:

1. Construct the periodic graph with one directed minimum-image displacement and integer image vector for each bond, then enumerate the ice basis from the tetrahedron constraints.

2. From each source ice state, apply all allowed directed exchanges. Continue a row while it lies outside $\mathcal{I}$, multiply by the exact resolvent denominator of every intermediate state, and record every return to $\mathcal{I}$ through the chosen order.

3. Before rows with the same source and target are combined, evaluate $\boldsymbol{\delta}_r$ from Eq. (15). Retain only $\boldsymbol{\delta}_r = 0$. For the second-/third-order FCC-32 calculation, applying Eq. (25) after combination is numerically identical.

4. Assemble the Hermitian projected matrix, diagonalize it once, and evaluate thermodynamic and spectral observables from that common eigensystem.

The construction recovers the leading bulk-topology effective Hamiltonian available on FCC-32; it does not turn FCC-32 into the thermodynamic limit. At higher order, the exact row filter remains valid, while a discrete character implementation must use enough points to resolve the largest transport represented.

THERMODYNAMIC AND SPECTRAL BENCHMARKS

The 2,970-dimensional periodic and projected matrices are diagonalized densely. For eigenvalues E_n,

$$C(T) = \frac{\langle E^2\rangle_T - \langle E\rangle_T^2}{T^2}. \tag{26}$$

The plotted heat capacity is C/N. For $|J_\pm|/J_{zz} = 0.03, 0.04, 0.05, 0.06, 0.08, 0.10$, the projected operator is strictly proportional to g_{hex} and its full spectrum collapses in units of Eq. (9). Its heat-capacity maximum is

$$T_{\text{peak}}/g_{\text{hex}} = 0.5210289229. \tag{27}$$

The periodic operator also contains g_4 and therefore cannot obey this cubic scaling. This simultaneously tests that the projection removes the lower-order artifact and leaves the physical ring matrix unchanged.

For either Hamiltonian, Letter Fig. 3 uses the four-sublattice trace of the longitudinal pseudospin correlation,

$$S^{zz}(\boldsymbol{q},\omega) = \frac{1}{Nd_0}\sum_{g\in\mathcal{G}}\sum_{n\notin\mathcal{G}}\sum_{a=0}^{3}\left|\sum_{j\in a} e^{-2\pi i\boldsymbol{q}\cdot\boldsymbol{r}_j}\langle n|S_j^z|g\rangle\right|^2 \delta[\omega-(E_n-E_g)]. \tag{28}$$

Here $\mathcal{G}$ is the numerically degenerate ground multiplet, d_0 is its dimension, and a labels the pyrochlore sublattice. Equation (28) contains no local-axis weights, magnetic form factor, or laboratory neutron-polarization tensor. The fully coherent source obtained by summing the four sublattice amplitudes before taking the modulus vanishes at the exact FCC-32 momenta by the ice constraint; the sublattice trace remains finite. Delta functions are represented by Gaussians of width $0.07g_{\text{hex}}$, and the frequency axis is linear.

The FCC-32 translation group has exactly eight characters. In conventional reciprocal-lattice units, and in the exact order plotted, they are

$$\begin{aligned}&(0,0,0),\ (-\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}),(\tfrac{1}{2},-\tfrac{1}{2},\tfrac{1}{2}),\ (0,0,1),\\ &(\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2}),\ (0,1,0),\ (1,0,0),(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}).\end{aligned} \tag{29}$$

These are Γ, four L representatives, and three X representatives. Letter Fig. 3 uses only Eq. (29); no smooth momentum path or fictitious finite-cluster momentum is shown.

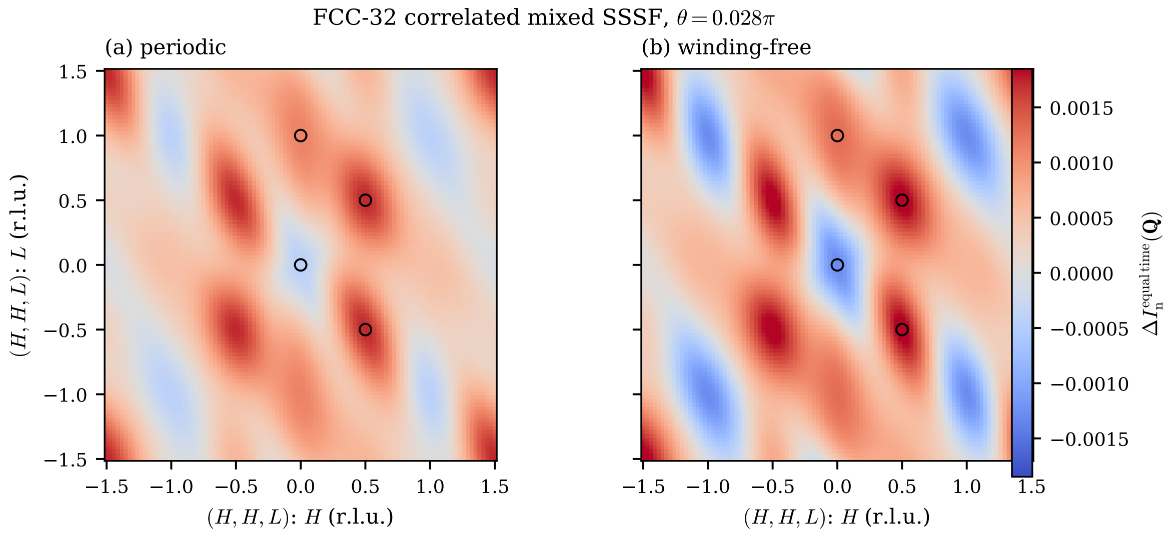

FIG. 1. FCC-32 global unpolarized equal-time structure factor in the (H, H, L) plane, with the onsite 1/6 term subtracted. Open circles mark the representatives in this plane of the eight independent momenta in Eq. (29). The continuous map is an extended-zone form-factor evaluation of the finite 32×32 correlation matrix, not a continuum of independent many-body momentum sectors.

GLOBAL NEUTRON PROJECTION AND EQUAL-TIME CHECK

The S^{zz} response in Eq. (28) deliberately omits physical single-crystal geometry. For the separate equal-time check below, a neutron probes the dipole in Eq. (13), projected from each local axis into the laboratory frame. For the four pyrochlore sublattices,

$$\hat{\boldsymbol{z}}_0 = \frac{(1,1,1)}{\sqrt{3}}, \quad \hat{\boldsymbol{z}}_1 = \frac{(1,-1,-1)}{\sqrt{3}}, \quad \hat{\boldsymbol{z}}_2 = \frac{(-1,1,-1)}{\sqrt{3}}, \quad \hat{\boldsymbol{z}}_3 = \frac{(-1,-1,1)}{\sqrt{3}}. \tag{30}$$

The unpolarized projector is $P_{ab}(\hat{\boldsymbol{Q}}) = \delta_{ab} - \hat{Q}_a \hat{Q}_b$, giving the site-pair factor

$$K_{ij}(\boldsymbol{Q}) = \hat{\boldsymbol{z}}_i \cdot \hat{\boldsymbol{z}}_j - (\hat{\boldsymbol{Q}} \cdot \hat{\boldsymbol{z}}_i)(\hat{\boldsymbol{Q}} \cdot \hat{\boldsymbol{z}}_j). \tag{31}$$

The local axes must remain inside both site sums; replacing K_{ij} by a single powder factor removes sublattice interference.

Longitudinal–transverse cross correlators vanish in the $U(1)$ reduction because S^z preserves total S^z while $S^\pm$ changes it. The physical unpolarized response therefore has the channel decomposition

$$I_{\rm mix}(\boldsymbol{Q}, \omega) = \sin^2\theta\, I_{zz}(\boldsymbol{Q}, \omega) + \cos^2\theta\, I_{+-}(\boldsymbol{Q}, \omega), \tag{32}$$

with $K_{ij}(\boldsymbol{Q})$ in each site-resolved correlator. Since $PS_i^\pm P = 0$, the dynamical transverse channel requires states outside the ice manifold and is not included in Letter Fig. 3. This restriction avoids representing spinon excitations by an invalid ice-only operator.

As a geometry and source-normalization check, we evaluate the equal-time version of Eq. (32) directly on FCC-32. The transverse equal-time contribution is the exact source norm $\langle S^-_{-\boldsymbol{Q}} S^+_{\boldsymbol{Q}} \rangle/2$ and does not require diagonalization of the unrestricted one-spinon sector. Figure 1 uses $|\theta| = 0.028\pi$ and subtracts the momentum-independent spin-1/2 onsite term 1/6 to expose the correlated signal.

$\mathrm{CE_2HF_2O_7}$ CONSTRAINT

Smith *et al.* report heat-capacity maxima near 0.065 and 0.025 K and give the seventh-order NLC parameter set A [1]

$$(J_a, J_b, J_c)_A = (0.050, 0.021, 0.004) \text{ meV}. \tag{33}$$

Taking J_a as the dominant gauge axis and using Eq. (12) gives

$$J_{zz} = 0.050 \text{ meV}, \qquad J_{\pm} = -0.00625 \text{ meV}. \tag{34}$$

Together with Eq. (27), this predicts $T_{\text{ring}} = 7.1$ mK, well below the lower observed feature.

The candidate A* in Letter Fig. 4 is defined by three equations, not by fitting digitized points:

$$\sum_\alpha (J_\alpha^\star)^2 = \sum_\alpha (J_\alpha^A)^2, \tag{35}$$

$$J_b^\star - J_c^\star = J_b^A - J_c^A, \tag{36}$$

$$\frac{0.5210289229}{k_B} \frac{12|(J_b^\star + J_c^\star)/4|^3}{(J_a^\star)^2} = 0.025 \text{ K}. \tag{37}$$

The solution is

$$(J_a, J_b, J_c)_{A^\star} = (0.0464354, 0.0266142, 0.0096142) \text{ meV}, \tag{38}$$

or $J_\pm^\star = -0.0090571$ meV. The first equation preserves the leading high-temperature second moment, while the second keeps the transverse anisotropy of A. These constraints make A* a falsifiable target for a new seventh-order NLC calculation. They do not prove that its complete high-temperature line shape fits the experiment.

The experimental points and published NLC-A curve in Letter Fig. 4 were digitized from Fig. 2(a) of Ref. [1], using the source raster and its logarithmic temperature calibration. They are included only for visual comparison because the source does not provide tabulated uncertainties. No likelihood or fit statistic is assigned to those digitized data.

REPRODUCIBILITY

The submission package contains the complete FCC-32 numerical record used in the Letter:

- `scripts/fcc32_prl_figures.py` constructs the cluster, enumerates second- and third-order rows, verifies the mask, diagonalizes the periodic and projected operators, and generates Letter Figs. 1–4;
- `data/fcc32_prl_results.npz` contains the spectra, thermodynamic peaks, allowed momenta, and DSSF arrays;
- `data/fcc32_prl_results.json` records dimensions, matrix-entry counts, the exact mask residual, and the A* constraints;
- `scripts/digitize_smith2025_heat_capacity.py` documents the figure calibration used to create `data/ce2hf2o7_smith2025_digitized.csv`;
- `data/fcc32_mixed_sssf.npz` contains the equal-time global neutron-projection check in Fig. 1; and
- `scripts/recompute_finite_size_artifact.py` is the vendored FCC geometry, row-enumeration, projection, and thermodynamics module.

All frequencies in Letter Fig. 3 are linear, and all of its columns are the exact FCC-32 momenta listed in Eq. (29). Generated arrays are stored separately from plotting code to permit direct numerical regression.

[1] E. M. Smith, A. Fitterman, R. Schäfer, *et al.*, Two-peak heat capacity accounts for $R\ln(2)$ entropy and ground state access in the dipole-octupole pyrochlore Ce2Hf2O7, Phys. Rev. Lett. **135**, 086702 (2025).